OFFICIAL DOCUMENTATION

3. Requirements Gathering

3.1 Stakeholder Analysis

- **End Users (Students & Learners):** Require a mobile platform that translates long-term goals into actionable daily plans. They are highly motivated by gamification (XP, badges), social challenges, and real-time push notifications.
- **System Administrators & Developers:** Focus on building and maintaining a scalable, serverless architecture on AWS. They prioritize system observability via CloudWatch, modular design, and robust security.

3.2 User Stories & Use Cases

- **Goal & Plan Generation:** As a user, I want to define my goal and upload personal learning materials, so the AI can generate a structured, context-aware study plan.
- **Grounded Learning (RAG):** As a user, I want my uploaded S3 documents to be securely indexed, so the AI can generate accurate quizzes and reviews based *only* on my materials.
- **Server-Side Gamification:** As a user, I want my streaks, XP, and badges calculated automatically upon task completion to ensure a fair and tamper-proof progression system.
- **Push Notifications:** As a user, I want to receive mobile push notifications reminding me of incomplete daily tasks before my deadline.

3.3 Functional Requirements (MVP)

ID	Feature Category	Description
FR-01	Auth	Users can register, sign in, sign out, and recover access via Cognito.
FR-02 / 03	AI Planning	Users create goals with time budgets; system generates multi-day plans.
FR-04 / 05	Task Mgmt	Users view daily tasks and mark them as completed idempotently.
FR-06 / 07	Streaks	Backend natively updates streak counters and consumes protections.
FR-10 / 11	S3 & RAG	Users upload private files to S3; system indexes them in Knowledge Bases.

3.4 Non-Functional Requirements

Requirement	Architectural Constraint & Criteria
Security	Enforce Amazon Cognito authentication, least-privilege IAM roles, private S3 objects, controlled presigned access, and strict server-side ownership validation for all resources.
Scalability	Utilize fully managed/serverless AWS services (API Gateway, Lambda, DynamoDB) to seamlessly handle bursty API traffic, scheduled events, and asynchronous AI workloads.
Reliability	Design completion and gamification operations to be completely idempotent. Implement automatic retries for asynchronous jobs (e.g., S3 indexing, push notification delivery).
Observability	Maintain comprehensive CloudWatch logs, operational metrics, custom alarms, and cross-service correlation identifiers for rapid troubleshooting.
Privacy	Ensure user-uploaded learning files remain strictly private. Document access must be strictly scoped to the authenticated owner's ID.
Maintainability	Maintain strict separation between mobile, API, domain, data-access, AI/RAG, notification, and scheduled-job modules to support future expansion.

Critical Constraint: Grounding Quality

AI outputs generated from retrieved document context must retain source metadata wherever practical. **The system must not silently treat missing context as factual content.** If relevant material cannot be retrieved from the Knowledge Base, the system must return a controlled 422 `Insufficient source context` error rather than hallucinating answers.